Week 4 · Part E — justification for the Monday review figure (partD_figure.png)

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1 · Each panel: its question, why that chart type, which Week 2 channel rule

Panel	The question it answers (one sentence)	Why this chart type	Week 2 channel rule
1 · daily sessions + 7-day mean	Where is traffic heading once the weekend rhythm is removed?	A line, because time is on the x axis and equal distance must mean equal duration. Raw kept faint (smoothing moves noise, it does not remove it); the window is named in the title.	Position on a common scale, plus **one pop-out**: raw in grey for context, the signal in colour.
2 · day × hour heatmap	When in the week and day is traffic highest?	168 cells of one measure over two cyclic categories; the pattern (weekends run 35% above weekdays, evening peak, busiest Sat 20:00) is the point, not a value. One-hue sequential Blues, no rainbow, no midpoint.	Position carries the two dimensions; colour lightness (a weaker channel) carries the value, which is acceptable for finding a pattern, not for reading amounts.
3 · treemap	Where is the mass in the catalogue — which categories and sub-categories dominate?	A part-to-whole of a *hierarchy*: containment shows parent → child and area shows share. One hue per parent; values printed on blocks.	**Area** is the fourth-best channel (±20%), so anything that matters carries its number in text. It answers “where”, not “by how much”.
4 · sorted bars	By how much do the sub-categories differ, and in what order (Phones lead Laptops by 6.5 points)?	The same eight numbers where the reader must *compare close values*: sorted horizontal bars, zero baseline, values printed. Hue matches the treemap so the two panels link.	**Position on a common scale**, the best channel — the one that treemap area cannot offer.

Palette (proved with vizlib.worst_pair, every pair): worst OKLab ΔE ×100 is 16.4 normal, 7.5 deutan, 9.5 protan, 8.4 tritan. The 4th hue (#CC79A7, Computing) is thin under deutan and below 3:1 contrast, so every parent is also named in text. Aspect ratio of panel 1 stated on the figure (1.74:1).

2 · Every time decision, named and defended

Parse. Two formats: ISO (9,019 rows) and DD/MM/YYYY (3,651 rows). The ambiguous one is resolved by evidence: the first field exceeds 12 on 2,151 rows and the second on 0, so it is day-first; 1,500 rows have both fields ≤12 and are decided only by that rule. *format='mixed'* parses every row, raises nothing, and moves **1,382 rows** to the wrong date (index 2024-01-04 → 2025-12-08, not 2024-04-01 → 2025-09-11); I checked the index min/max, not the row count. **Resolution/aggregation.** Hourly flows → daily with **sum** (axis says “sum of 24 hourly counts”); a flow is never resampled with last(). Gaps were counted on the hourly index (312 h = 13 whole days) *before* resampling and left as breaks, not zeros or interpolation. **Window.** 7 days, because the cycle removed is weekly and 7 is the shortest window that cancels it exactly (Part B.1: weekly swing left 0.2%); longer windows start eating the seasonal turn (91 d keeps 50% of the wave and loses 17% of the series to the ends). **Centred**, because the Monday review is retrospective and all data exist; a live dashboard must use a trailing mean and say so (Part B.4: lag = half-window).

The window clause. The window was fixed at 7 days before plotting, because the cycle being removed is weekly (Part B.1). The series is shown from its first recorded day; no start date was selected. A 28-day version is in the appendix (partD_appendix_28day.png) and does not change the direction — checked, not asserted: both windows put the spring trough in late spring and show a rise into September (+8,914 vs +7,339 sessions/day). Part B.5 shows why this matters: the same series gives 34 rising and 26 falling trends across 60 start/window choices. **Symmetry test:** had the 7-day window made the September rise look smaller, I would still have used it, because I chose it for the weekly cycle and published the 28-day version anyway; I would show the same figure if it weakened my conclusion.

3 · Every row removed or altered — rule, count, from → to

Rule	Count	What moved (from → to)
Exact duplicate hourly rows (same hour, same values; many written in the other date format) — dropped, first kept	286	12,670 -> 12,384 rows; total valid sessions 12,010,834 -> 11,727,160 (-2.36%)
sessions = −1 (no-reading sentinel) — set to NaN, not 0; row kept, orders/revenue kept	287	mean sessions/hour 946.9 → 969.4; 222 days topped up from the same-weekday profile (mean +5.8%, max +19.8%), run both ways in Part B (no conclusion changes)
Never-recorded days — NOT removed, NOT filled: left as gaps	13 days (312 h)	529 calendar days → 516 observed; marked on the figure
Device spellings normalised (strip, lower, Title)	1,447 relabelled	6 spellings → 3 (Desktop, Mobile, Tablet); none removed
Duplicate SKU: SKU-0127 (AUDI-04) listed under Audio *and* Wearables — Wearables copy removed (a tree needs one parent)	1	345 → 344 rows; Wearables share 10.13% → 9.35%
Negative net revenue — no area, so removed from the treemap/bars and reported here	7 SKUs (-1.07 PKR m)	344 → 337 SKUs; catalogue net 1,711.4 → 1,712.5 PKR m
Partial final month (September 2025, 11 of 30 days) — kept, shaded and labelled; never compared as a month	n/a	1–11 Sep 2025 compared only with 1–11 Sep 2024 (+37%)

After cleaning, the eight sub-category shares land within 0.23 points of shares_eight.csv (limit 0.3). Before the duplicate-SKU fix Wearables was out by 0.93 points.

4 · Every level rolled up

Other bucket: none. tslib.rollup_other at 1% reports “no categories below 1.0%” — the smallest sub-category (Drones) is 2.4%. **Depth cut:** the tree is category → sub-category; the 337 SKU leaves are rolled into 8 sub-categories (4 parents, Mobile 38% / Audio & Imaging 23% / Lifestyle 20% / Computing 18%) because a 337-block treemap is unreadable and no reader question needs SKU detail. 8 leaves is within a treemap's range but above the ~4±1 chunks of working memory, which is why panel 4 exists. Checks: children sum to parent, shares sum to 100% (asserted in code), no negative values drawn.